

1) Find all solutions of given equation.

$$\sqrt{2} \sin \theta + 1 = 0$$

$$\sqrt{2} \sin \theta = -1$$

$$\sin \theta = -\frac{\sqrt{2}}{2}$$

$$\theta = \sin^{-1}\left(-\frac{\sqrt{2}}{2}\right)$$

$$Q3: \theta_1 = \frac{5\pi}{4} + 2\pi k$$

$$Q4: \theta_2 = \frac{7\pi}{4} + 2\pi k$$

$$\theta = \frac{5\pi}{4} + 2\pi k, \frac{7\pi}{4} + 2\pi k$$

2) Find all solutions of the given equation.

$$\sec^2 \theta - 4 = 0$$

$$\sec^2 \theta = 4$$

$$\sec \theta = \pm \sqrt{4}$$

$$\sec \theta = 2 \quad \text{or}$$

$$\sec \theta = -2$$

$$\cos \theta = \frac{1}{2}$$

$$\cos \theta = -\frac{1}{2}$$

$$Q1: \theta = \frac{\pi}{3} + 2\pi k$$

$$Q3: \theta = \frac{4\pi}{3} + 2\pi k$$

$$Q2: \theta = \frac{2\pi}{3} + 2\pi k$$

$$Q4: \theta = \frac{5\pi}{3} + 2\pi k$$

$$\theta = \frac{\pi}{3} + 2\pi k, \frac{2\pi}{3} + 2\pi k, \frac{4\pi}{3} + 2\pi k, \frac{5\pi}{3} + 2\pi k$$

3) Find all solutions of given equation.

$$\csc^2 \theta - 2 = 0$$

$$\csc^2 \theta = 2$$

$$\csc \theta = \pm \sqrt{2}$$

$$\csc \theta = \sqrt{2} \quad \text{or}$$

$$\csc \theta = -\sqrt{2}$$

$$\sin \theta = \frac{\sqrt{2}}{2}$$

$$\sin \theta = -\frac{\sqrt{2}}{2}$$

$$Q1: \theta = \frac{\pi}{4} + 2\pi k$$

$$Q3: \theta = \frac{5\pi}{4} + 2\pi k$$

$$Q2: \theta = \frac{3\pi}{4} + 2\pi k$$

$$Q4: \theta = \frac{7\pi}{4} + 2\pi k$$

$$\theta = \frac{\pi}{4} + 2\pi k, \frac{3\pi}{4} + 2\pi k, \frac{5\pi}{4} + 2\pi k, \frac{7\pi}{4} + 2\pi k$$

4) Solve the given equation.

$$4\sin^2 \theta - 4\sin \theta + 1 = 0$$

$$\text{Let } u = \sin \theta$$

$$4u^2 - 4u + 1 = 0$$

$$(2u-1)^2 = 0$$

$$2u-1=0$$

$$u = \frac{1}{2}$$

$$\sin \theta = \frac{1}{2}$$

$$Q1: \theta = \frac{\pi}{6} + 2\pi k$$

$$Q2: \theta = \frac{5\pi}{6} + 2\pi k$$

$$\theta = \frac{\pi}{6} + 2\pi k, \frac{5\pi}{6} + 2\pi k$$

5) Solve the given equation.

$$2\cos^2 \theta - \cos \theta - 1 = 0$$

$$\text{Let } u = \cos \theta$$

$$2u^2 - u - 1 = 0$$

$$2u^2 - 2u + u - 1 = 0$$

$$2u(u-1) + (u-1) = 0$$

$$(2u+1)(u-1) = 0$$

$$2u+1=0 \quad \text{or}$$

$$u-1=0$$

$$u = -\frac{1}{2}$$

$$u = 1$$

$$\cos \theta = -\frac{1}{2}$$

$$\cos \theta = 1$$

$$Q2: \frac{2\pi}{3} + 2\pi k$$

$$\theta = 0 + 2\pi k$$

$$Q3: \frac{4\pi}{3} + 2\pi k$$

$$\theta = \frac{2\pi}{3} + 2\pi k, \frac{4\pi}{3} + 2\pi k, 2\pi k$$

6) Solve the given equation

$$3 + \tan^3 \theta = \tan \theta$$

$$3 + \tan^3 \theta - \tan \theta = 0$$

$$\tan \theta (3 + \tan^2 \theta - 1) = 0$$

$$\tan \theta = 0 \quad \text{or} \quad 3 + \tan^2 \theta - 1 = 0$$

$$\theta = 0 + \pi k$$

$$\tan \theta = \pm \frac{1}{\sqrt{3}}$$

$$\tan \theta = \frac{1}{\sqrt{3}} \quad \text{or} \quad \tan \theta = -\frac{1}{\sqrt{3}}$$

$$\theta = \frac{\pi}{6} + \pi k$$

$$\theta = -\frac{\pi}{6} + \pi k$$

$$\theta = \pi k, \frac{\pi}{6} + \pi k, -\frac{\pi}{6} + \pi k$$

7) Solve the given equation and list six specific solutions.

$$\cos \theta = \frac{1}{2}$$

$$Q1: \theta = \frac{\pi}{3} + 2\pi k$$

$$Q4: \theta = \frac{5\pi}{3} + 2\pi k$$

$$\theta = \frac{\pi}{3} + 2\pi k, \frac{5\pi}{3} + 2\pi k$$

$$k=0: \theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$k=1: \theta = \frac{7\pi}{3}, \frac{11\pi}{3}$$

$$k=2: \theta = \frac{13\pi}{3}, \frac{17\pi}{3}$$

$$\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, \frac{13\pi}{3}, \frac{17\pi}{3}$$

8) Solve the given equation and list six specific solutions.

$$\tan \theta = -11$$

$$\theta = \tan^{-1}(-11)$$

$$\theta \approx -1.48$$

$$\theta = -1.48 + \pi k$$

$$k=0: \theta = -1.48$$

$$k=1: \theta = 1.66$$

$$k=2: \theta = 4.80$$

$$k=3: \theta = 7.94$$

$$k=4: \theta = 11.09$$

$$k=5: \theta = 14.23$$

$$\theta = -1.48, 1.66, 4.80, 7.94, 11.09, 14.23$$

9) Solve the given equation

$$3\sin^2 \theta + 13\sin \theta - 30 = 0$$

$$\text{Let } u = \sin \theta$$

$$3u^2 + 13u - 30 = 0$$

$$3u^2 + 18u - 5u - 30 = 0$$

$$3u(u+6) - 5(u+6) = 0$$

$$(3u-5)(u+6) = 0$$

$$3u-5=0 \quad \text{or} \quad u+6=0$$

$$u = \frac{5}{3} \quad u = -6$$

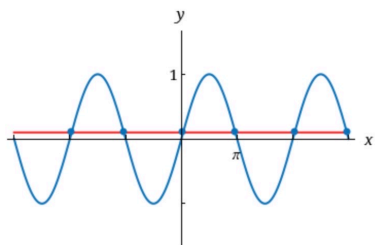
$$\sin \theta = \frac{5}{3} \quad \sin \theta = -6$$

$$\frac{5}{3} > 1 \quad -6 < -1$$

NO SOLUTION

10) We can find some of the solutions of $\sin(x) = 0.1$ graphically by graphing $y = \sin(x)$ and $y = 0.1$.

Use the graph below to estimate some of the solutions. (Let $-3\pi < x < 3\pi$. Enter your answers as a comma-separated list. Round your answers to four decimal places.)



$$x = -6.1830, -3.2418, 0.1002, 3.0414, 6.3834, 9.3246$$

Ⓢ

That's it!

11) Solve the given equation.

$$\cos \theta = 1$$

$$\theta = 0 + 2\pi k$$

$$\theta = 2\pi k$$

12) Solve the given equation and list six specific solutions.

$$\sin \theta = -0.6$$

$$\theta = \sin^{-1}(-0.6)$$

$$Q3: \theta = -0.64 + 2\pi k$$

$$Q4: \theta = 3.79 + 2\pi k$$

$$\theta = -0.64 + 2\pi k, 3.79 + 2\pi k$$

$$k=0: \theta = -0.64, 3.79$$

$$k=1: \theta = 5.64, 6.28$$

$$k=2: \theta = 11.92, 16.36$$

$$\theta = -0.64, 3.79, 5.64, 6.28, 11.92, 16.36$$

13) Find all solutions of given equation.

$$4\cos^2\theta - 3 = 0$$

$$\cos\theta = \pm \frac{\sqrt{3}}{2}$$

$$\cos\theta = \frac{\sqrt{3}}{2} \quad \text{or} \quad \cos\theta = -\frac{\sqrt{3}}{2}$$

$$Q1: \theta = \frac{\pi}{6} + 2\pi k \quad Q2: \theta = \frac{5\pi}{6} + 2\pi k$$

$$Q4: \theta = \frac{11\pi}{6} + 2\pi k \quad Q3: \theta = \frac{7\pi}{6} + 2\pi k$$

$$\theta = \frac{\pi}{6} + 2\pi k, \frac{11\pi}{6} + 2\pi k, \frac{5\pi}{6} + 2\pi k, \frac{7\pi}{6} + 2\pi k$$

14) Solve the given equation.

$$(\tan^2\theta - 16)(2\cos\theta + 1) = 0$$

$$\tan^2\theta - 16 = 0$$

$$\text{or} \quad 2\cos\theta + 1 = 0$$

$$(\tan\theta + 4)(\tan\theta - 4) = 0$$

$$\cos\theta = -\frac{1}{2}$$

$$\tan\theta + 4 = 0 \quad \text{or} \quad \tan\theta - 4 = 0$$

$$\tan\theta = -4 \quad \tan\theta = 4$$

$$Q2: \theta = \frac{2\pi}{3} + 2\pi k$$

$$\theta = -1.326 + \pi k \quad \theta = 1.326 + \pi k$$

$$Q3: \theta = \frac{4\pi}{3} + 2\pi k$$

$$\theta = -1.326 + \pi k, 1.326 + \pi k, \frac{2\pi}{3} + 2\pi k, \frac{4\pi}{3} + 2\pi k$$

7.5

1) Solve the given equation.

$$\cos 2\theta = \cos^2\theta - \frac{1}{2}$$

$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

$$\sin^2\theta = \frac{1}{2}$$

$$\sin\theta = \pm \frac{\sqrt{2}}{2}$$

$$\sin\theta = \frac{\sqrt{2}}{2} \quad \text{or} \quad \sin\theta = -\frac{\sqrt{2}}{2}$$

$$Q1: \theta = \frac{\pi}{4} + 2\pi k$$

$$Q3: \theta = \frac{5\pi}{4} + 2\pi k$$

$$Q2: \theta = \frac{3\pi}{4} + 2\pi k$$

$$Q4: \theta = \frac{7\pi}{4} + 2\pi k$$

$$\theta = \frac{\pi}{4} + 2\pi k, \frac{3\pi}{4} + 2\pi k, \frac{5\pi}{4} + 2\pi k, \frac{7\pi}{4} + 2\pi k$$

2) Solve the given equation.

$$\tan\theta - 3\cot\theta = 0$$

$$\tan\theta - 3\left(\frac{1}{\tan\theta}\right) = 0$$

$$\tan^2\theta - 3 = 0$$

$$\tan\theta = \pm\sqrt{3}$$

$$\tan\theta = \sqrt{3} \quad \text{or} \quad \tan\theta = -\sqrt{3}$$

$$\theta = \frac{\pi}{3} + \pi k \quad \theta = -\frac{\pi}{3} + \pi k$$

$$\theta = \frac{\pi}{3} + \pi k, -\frac{\pi}{3} + \pi k$$

3) An equation is given. Find all solutions of the equation and list solutions in the interval $[0, 2\pi)$

$$\sin\left(\frac{\theta}{2} - 1\right) = 0$$

$$\sin\left(\frac{\theta}{2}\right) = 1$$

$$\frac{\theta}{2} = \frac{\pi}{2} + 2\pi k$$

$$\theta = \pi + 4\pi k$$

$$k=0: \theta = \pi$$

$$k=1: \theta = 5\pi$$

$$k=-1: \theta = -3\pi$$

$$\theta = \pi$$

4) An equation is given. Find all solutions of the equation and list solutions in the interval $[0, 2\pi)$

$$\csc\left(\frac{\theta}{2}\right) = \sin\left(\frac{\theta}{2}\right)$$

$$\frac{1}{\sin\left(\frac{\theta}{2}\right)} = \sin\left(\frac{\theta}{2}\right)$$

$$\sin^2\left(\frac{\theta}{2}\right) = 1$$

$$\sin\left(\frac{\theta}{2}\right) = \pm 1$$

$$\frac{\theta}{2} = \frac{\pi}{2} + \pi k$$

$$\theta = \pi + 2\pi k$$

$$k=0: \theta = \pi$$

$$k=1: \theta = 3\pi$$

$$k=-1: \theta = -\pi$$

$$\theta = \pi$$

5) Use Double- or Half-Angle Formula to solve the equation in the interval $[0, 2\pi)$.

$$\tan\left(\frac{\theta}{2}\right) - \sin\theta = 0$$

$$\frac{1 - \cos\theta}{\sin\theta} - \sin\theta = 0$$

$$\begin{aligned}
 1 - \cos \theta - \sin^2 \theta &= 0 \\
 1 - \cos \theta - (1 - \cos^2 \theta) &= 0 \\
 1 - \cos \theta - 1 + \cos^2 \theta &= 0 \\
 \cos^2 \theta - \cos \theta &= 0 \\
 \cos \theta (\cos \theta - 1) &= 0 \\
 \cos \theta = 0 &\quad \text{or} \quad \cos \theta - 1 = 0 \\
 \theta = \frac{\pi}{2} &\quad \cos \theta = 1 \\
 \theta = \frac{3\pi}{2} &\quad \theta = 0
 \end{aligned}$$

$$\theta = 0, \frac{\pi}{2}, \frac{3\pi}{2}$$

6) Use Double- or Half-Angle Formula to solve the equation in the interval $[0, 2\pi)$.

$$2 \sin^2 \theta = 2 + \cos 2\theta$$

$$2 \sin^2 \theta = 2 + 1 - 2 \sin^2 \theta$$

$$4 \sin^2 \theta = 3$$

$$\sin^2 \theta = \frac{3}{4}$$

$$\sin \theta = \pm \frac{\sqrt{3}}{2}$$

$$\sin \theta = \frac{\sqrt{3}}{2} \quad \text{or} \quad \sin \theta = -\frac{\sqrt{3}}{2}$$

$$Q1: \theta = \frac{\pi}{3} \quad Q3: \theta = \frac{4\pi}{3}$$

$$Q2: \theta = \frac{2\pi}{3} \quad Q4: \theta = \frac{5\pi}{3}$$

$$\theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$$

7) Use Double- or Half-Angle Formula to solve the equation in the interval $[0, 2\pi)$.

$$-\sin 2\theta - \cos 4\theta = 0$$

$$\cos 4\theta + \sin 2\theta = 0$$

$$(1 - 2 \sin^2 2\theta) + \sin 2\theta = 0$$

$$-2 \sin^2 2\theta + \sin 2\theta + 1 = 0$$

$$2 \sin^2 2\theta - \sin 2\theta - 1 = 0$$

$$\text{Let } u = \sin 2\theta$$

$$2u^2 - u - 1 = 0$$

$$2u^2 - 2u + u - 1 = 0$$

$$2u(u - 1) + (u - 1) = 0$$

$$(2u + 1)(u - 1) = 0$$

$$2u + 1 = 0 \quad \text{or} \quad u - 1 = 0$$

$$u = -\frac{1}{2} \quad u = 1$$

$$\sin 2\theta = -\frac{1}{2} \quad \sin 2\theta = 1$$

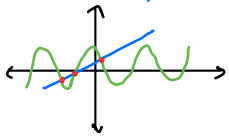
$$2\theta = \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{19\pi}{6}, \frac{23\pi}{6} \quad 2\theta = \frac{\pi}{2}, \frac{5\pi}{2}$$

$$\theta = \frac{7\pi}{12}, \frac{11\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12} \quad \theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

$$\theta = \frac{7\pi}{12}, \frac{11\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}, \frac{\pi}{4}, \frac{5\pi}{4}$$

8) Using graphing calculator to find solutions of the equation.

$$\cos(x) = \frac{x}{4}$$



$$x = -3.60, -2.13, 1.25$$

9) Displacement of a spring vibrating in damped harmonic motion is given by following equation. Find first 4 times when spring is at its equilibrium position $y=0$.

$$y = 4e^{-9t} \sin(2\pi t)$$

$$0 = 4e^{-9t} \sin(2\pi t)$$

$$4 \neq 0$$

$$e^{-9t} \neq 0$$

$$\sin(2\pi t) = 0$$

$$2\pi t \text{ should be multiple of } \pi$$

$$t = 0, \frac{1}{2}, 1, \frac{3}{2}$$

10) An equation is given. Find all solutions of the equation and list solutions in the interval $[0, 2\pi)$.

$$\sin 2\theta = 8 \cos 2\theta$$

$$\frac{\sin 2\theta}{\cos 2\theta} = 8$$

$$\tan 2\theta = 8$$

$$2\theta = \tan^{-1}(8) + k\pi$$

$$\theta = \frac{\tan^{-1}(8)}{2} + \frac{k\pi}{2}$$

$$\theta = 0.723 + \frac{k\pi}{2}$$

$$k=0: \theta = \frac{\tan^{-1}(8)}{2} \approx 0.723$$

$$k=1: \theta = \frac{\tan^{-1}(8)}{2} + \frac{\pi}{2} \approx 2.294$$

$$k=2: \theta = \frac{\tan^{-1}(8)}{2} + \pi \approx 3.865$$

$$k=3: \theta = \frac{\tan^{-1}(8)}{2} + \frac{3\pi}{2} \approx 5.436$$

$$k=4: \theta = \frac{\tan^{-1}(8)}{2} + 2\pi \approx 7.006$$

$$\theta = 0.723, 2.294, 3.965, 5.436$$

- 11) An equation is given. Find all solutions of the equation and list solutions in the interval $[0, 2\pi)$

$$2\cos\left(\frac{\theta}{3}\right) + \sqrt{3} = 0$$

$$\cos\left(\frac{\theta}{3}\right) = -\frac{\sqrt{3}}{2}$$

$$Q2: \frac{\theta}{3} = \frac{5\pi}{6} + 2\pi k$$

$$\theta = \frac{5\pi}{2} + 6\pi k$$

$$Q3: \frac{\theta}{3} = \frac{7\pi}{6} + 2\pi k$$

$$\theta = \frac{7\pi}{2} + 6\pi k$$

$$\theta = \frac{5\pi}{2} + 6\pi k, \frac{7\pi}{2} + 6\pi k$$

$$k=0: \theta = \frac{5\pi}{2}$$

$$k=1: \theta = \frac{5\pi}{2} + 6\pi$$

$$k=-1: \theta = \frac{5\pi}{2} - 6\pi$$

NO SOLUTION

- 12) Solve the given equation.

$$3\sin(2\theta) - 2\sin\theta = 0$$

$$3(2\sin\theta\cos\theta) - 2\sin\theta = 0$$

$$6\sin\theta\cos\theta - 2\sin\theta = 0$$

$$2\sin\theta(3\cos\theta - 1) = 0$$

$$2\sin\theta = 0 \quad \text{or} \quad 3\cos\theta - 1 = 0$$

$$\sin\theta = 0 \quad \cos\theta = \frac{1}{3}$$

$$\theta = \pi k \quad Q1: \theta = 1.231 + 2\pi k$$

$$Q4: \theta = -1.231 + 2\pi k$$

$$\theta = \pi k, 1.231 + 2\pi k, -1.231 + 2\pi k$$

- 13) An equation is given. Find all solutions of the equation and list solutions in the interval $[0, 2\pi)$

$$2\sqrt{3}\tan(3\theta) - 2 = 0$$

$$\tan(3\theta) = \frac{1}{\sqrt{3}}$$

$$3\theta = \frac{\pi}{6} + \pi k$$

$$\theta = \frac{\pi}{18} + \frac{\pi k}{3}$$

$$k=0: \theta = \frac{\pi}{18}$$

$$k=1: \theta = \frac{7\pi}{18}$$

$$k=2: \theta = \frac{13\pi}{18}$$

$$k=3: \theta = \frac{19\pi}{18}$$

$$k=4: \theta = \frac{25\pi}{18}$$

$$k=5: \theta = \frac{31\pi}{18}$$

$$k=6: \theta = \frac{37\pi}{18}$$

$$\theta = \frac{\pi}{18}, \frac{7\pi}{18}, \frac{13\pi}{18}, \frac{19\pi}{18}, \frac{25\pi}{18}, \frac{31\pi}{18}$$

- 14) An equation is given. Find all solutions of the equation and list solutions in the interval $[0, 2\pi)$

$$\sec 3\theta - 2 = 0$$

$$\cos 3\theta = \frac{1}{2}$$

$$Q1: 3\theta = \frac{\pi}{3} + 2\pi k$$

$$\theta = \frac{\pi}{9} + \frac{2\pi k}{3}$$

$$Q4: 3\theta = \frac{5\pi}{3} + 2\pi k$$

$$\theta = \frac{5\pi}{9} + \frac{2\pi k}{3}$$

$$\theta = \frac{\pi}{9} + \frac{2\pi k}{3}, \frac{5\pi}{9} + \frac{2\pi k}{3}$$

$$k=0: \theta = \frac{\pi}{9}, \frac{5\pi}{9}$$

$$k=1: \theta = \frac{7\pi}{9}, \frac{11\pi}{9}$$

$$k=2: \theta = \frac{13\pi}{9}, \frac{17\pi}{9}$$

$$k=3: \theta = \frac{19\pi}{9}, \frac{23\pi}{9}$$

$$\theta = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}, \frac{11\pi}{9}, \frac{13\pi}{9}, \frac{17\pi}{9}$$

- 15) An equation is given. Solve the equation by factoring and list solutions in the interval $[0, 2\pi)$

$$\csc\theta \cot\theta - \sin\theta + \tan\theta = \cos\theta$$

$$\frac{1}{\sin\theta} \cdot \frac{\cos\theta}{\sin\theta} - \sin\theta \cdot \frac{\sin\theta}{\cos\theta} = \cos\theta$$

$$\frac{\cos\theta}{\sin^2\theta} - \frac{\sin^2\theta}{\cos\theta} = \cos\theta$$

$$\cos^2\theta - \sin^4\theta = \cos^2\theta \sin^2\theta$$

$$\cos^2\theta - \sin^4\theta - \cos^2\theta \sin^2\theta = 0$$

$$\cos^2\theta(1 - \sin^2\theta) - \sin^4\theta = 0$$

$$\begin{aligned}
 \cos^2 \theta \cdot \cos^2 \theta - \sin^4 \theta &= 0 \\
 \cos^4 \theta - \sin^4 \theta &= 0 \\
 (\cos^2 \theta - \sin^2 \theta)(\cos^2 \theta + \sin^2 \theta) &= 0 \\
 \cos^2 \theta - \sin^2 \theta &= 0 \\
 \cos 2\theta &= 0 \\
 2\theta &= \frac{\pi}{2} + \pi k \\
 \theta &= \frac{\pi}{4} + \frac{\pi k}{2}
 \end{aligned}$$

$$\begin{aligned}
 k=0: \theta &= \frac{\pi}{4} \\
 k=1: \theta &= \frac{3\pi}{4} \\
 k=2: \theta &= \frac{5\pi}{4} \\
 k=3: \theta &= \frac{7\pi}{4} \\
 k=4: \theta &= \frac{9\pi}{4}
 \end{aligned}$$

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

16) An equation is given. Solve the equation by factoring and list solutions in the interval $[0, 2\pi)$

$$\begin{aligned}
 2 \sin \theta \tan \theta + \tan \theta &= 1 + 2 \sin \theta \\
 2 \sin \theta \tan \theta + \tan \theta - 2 \sin \theta - 1 &= 0 \\
 \tan \theta (2 \sin \theta + 1) - (2 \sin \theta + 1) &= 0 \\
 (2 \sin \theta + 1)(\tan \theta - 1) &= 0 \\
 2 \sin \theta + 1 = 0 &\quad \text{or} \quad \tan \theta - 1 = 0 \\
 \sin \theta = -\frac{1}{2} &\quad \tan \theta = 1 \\
 Q3: \theta = \frac{7\pi}{6} + 2\pi k &\quad \theta = \frac{\pi}{4} + \pi k \\
 Q4: \theta = \frac{11\pi}{6} + 2\pi k
 \end{aligned}$$

$$\theta = \frac{7\pi}{6} + 2\pi k, \frac{11\pi}{6} + 2\pi k, \frac{\pi}{4} + \pi k$$

$$k=0: \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{\pi}{4}$$

$$k=1: \frac{5\pi}{4}$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{\pi}{4}, \frac{5\pi}{4}$$

17) Use Double- or Half-Angle Formula to solve the equation in the interval $[0, 2\pi)$.

$$\begin{aligned}
 \cos 2\theta + 7 \cos \theta &= 8 \\
 2 \cos^2 \theta - 1 + 7 \cos \theta &= 8 \\
 2 \cos^2 \theta + 7 \cos \theta - 9 &= 0 \\
 \text{Let } u &= \cos \theta \\
 2u^2 + 7u - 9 &= 0 \\
 2u^2 - 2u + 9u - 9 &= 0 \\
 2u(u-1) + 9(u-1) &= 0 \\
 (2u+9)(u-1) &= 0 \\
 2u+9=0 &\quad \text{or} \quad u-1=0 \\
 u = -\frac{9}{2} &\quad u=1 \\
 \cos \theta = -\frac{9}{2} &\quad \cos \theta = 1 \\
 -\frac{9}{2} < -1 &\quad \theta = 0
 \end{aligned}$$

$$\theta = 0$$

18) $2 \sin^{-1}(x) + \cos^{-1}(x) = \pi$

$$\begin{aligned}
 \cos^{-1}(x) &= \pi - 2 \sin^{-1}(x) \\
 \cos(\cos^{-1}(x)) &= \cos(\pi - 2 \sin^{-1}(x)) \\
 x &= -\cos(2 \sin^{-1}(x)) \\
 x &= -(1 - 2 \sin^2(\sin^{-1}(x))) \\
 x &= -(1 - 2x^2) \\
 x &= 2x^2 - 1 \\
 2x^2 - x - 1 &= 0 \\
 (2x+1)(x-1) &= 0 \\
 2x+1=0 &\quad \text{or} \quad x-1=0 \\
 x = -\frac{1}{2} &\quad x=1
 \end{aligned}$$